Friction resolution and deformation prediction

Secondary study appendix | Position-only calibration, unchanged coupled physics

Motivation and controlled change

A development object selected excessive stiffness when its friction lay between coarse candidates. A uniform nested friction grid tests whether this discretization causes compensating material estimates. This diagnosis motivated a secondary study; it does not establish that friction resolution is the only source of error.

The physical solver, predictor and reference meshes, timestep, contact, material candidates, observed positions and masks, noise realization, full-window calibration loss, cohort definitions and fixed videos are unchanged. No force observation is added. Reference and observation artifacts are reused read-only. The original coarse study is preserved.

Calibration family	Coarse candidates	Fine candidates
Nonlinear FEM	35 = 7 x 5	119 = 7 x 17
Adapted native spring core	35 = 7 x 5	119 = 7 x 17
Rigid contact baseline	5	17

The fine friction grid spans 0 to 0.8 in steps of 0.05. Of 255 candidates, 75 reuse the exact coarse outputs and 180 are additional. Greater search budget is part of this comparison; it is not an equal-compute architecture experiment.

Fixed reporting scope

All 40 development and reserved forecasts remain recorded, including failures. The reserved primary cohort contains all 24 slide, curved-obstacle and moving-body cases. Eight sharp-edge drops remain exploratory stress cases because development spatial qualification failed. Numerical qualification flags do not remove cases from either cohort. Means are conditional on completed predictions when coverage is incomplete.

Reserved prediction results

Method / search	Complete /24	World mm	Shape mm
FEM, coarse grid	24/24	5.155	0.973
FEM, fine grid	24/24	4.134	0.897
Adapted spring core, fine	16/24	8.704	0.970
Rigid contact, fine	24/24	6.743	0.975
True-parameter coarse FEM	24/24	3.338	0.837
Position ridge	24/24	45.466	4.414

World values are coordinate RMSE; shape values remove best-fitting rigid motion. They measure simulation forecast error, not calibrated physical uncertainty. The oracle uses true parameters in the coarse predictor, so its residual includes discretization error.

Fine FEM versus	Matched /24	Mean gain mm	Wins / losses / ties
Coarse FEM	24/24	1.021	13 / 5 / 6
Fine rigid	24/24	2.610	21 / 3 / 0
Fine spring core	16/24	4.366	16 / 0 / 0
Oracle	24/24	-0.796	7 / 17 / 0

Positive paired gain favors fine FEM. Comparisons use only jointly completed cases, with matched counts shown; failures are not imputed as favorable errors. No significance or population generalization is inferred from these descriptive means.

Stress and development accounting

Sharp-edge stress: fine FEM completes 8/8; conditional world / shape means 24.161 / 2.186 mm. Fine-versus-coarse paired gain is 3.448 mm over 8/8 cases. These targets remain development-unqualified.

Development primary regression check: 6/6 matched cases; mean fine-grid gain 11.319 mm; 4 improvements, 2 regressions and 0 ties. Development cases motivated this secondary study and are not independent confirmation.

Physical estimates, coverage and evidence

Object / split	True E / mu	Coarse FEM E / mu	Fine FEM E / mu
surface_dev0 / dev	6000 / 0.35	6500 / 0.40	6500 / 0.45
surface_dev1 / dev	3000 / 0.10	9000 / 0.20	3000 / 0.10
surface_test0 / reserved	4000 / 0.25	4500 / 0.40	4500 / 0.30
surface_test1 / reserved	4000 / 0.50	4500 / 0.60	4500 / 0.60
surface_test2 / reserved	7000 / 0.25	13000 / 0.40	6500 / 0.30
surface_test3 / reserved	7000 / 0.50	6500 / 0.60	6500 / 0.65
surface_test4 / reserved	10000 / 0.25	18000 / 0.40	13000 / 0.30
surface_test5 / reserved	10000 / 0.50	9000 / 0.60	9000 / 0.60
surface_test6 / reserved	14000 / 0.25	6500 / 0.20	18000 / 0.30
surface_test7 / reserved	14000 / 0.50	13000 / 0.60	13000 / 0.55

E is in Pa. Truth appears only in post-fit diagnostics, not candidate selection. Point estimates do not demonstrate identifiability or calibrated uncertainty. Native spring scale is not Young modulus.

Completed calibration candidates: fem 119/119, native 5/119, rigid 17/17. All guarded failures remain in the independent audit. The native comparison adapts the authors' spring-force core into shared contact and integration; it is not a full PhysTwin system reproduction. There is no matched fine-grid vertex-contact baseline.

Fixed actual-versus-predicted videos

surface_test0-curved.mp4 | primary

surface_test3-moving_sphere.mp4 | primary

surface_test7-edge_drop.mp4 | stress

The same three predeclared cases include the sharp-edge stress example. Playback is quarter speed with shared scale and camera. A clip is illustrative and does not replace full-cohort accounting. This secondary simulation study makes no official benchmark improvement, calibrated boundary-inference, active-probing advantage or general contact-solver convergence claim.